# Decentralized Crypto World Bank

A Blockchain-Based Hierarchical Lending Platform
with AI-Enhanced Security

by

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A pre-thesis 1 report submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
**B.Sc. in Computer Science**

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## Declaration

It is hereby declared that

1. The report submitted is my/our own original work while completing degree at Brac University.
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4. We have acknowledged all main sources of help.

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## Abstract

International development finance is delivered through layered institutional structures in which multilateral development institutions, national financial intermediaries, and local lenders jointly channel capital to end borrowers. While this model enables risk distribution and local market access, it is associated with operational challenges, including limited cross-tier transparency, lengthy cross-border settlement processes, high reconciliation and compliance overhead, and constrained scalability of risk assessment workflows. At the same time, approximately 1.4 billion adults remain unbanked according to the World Bank Global Findex (2021), highlighting the need for more efficient and accessible credit delivery mechanisms.

This paper presents the **Crypto World Bank**, a prototype decentralized application (DApp) that implements a four-tier hierarchical lending architecture on an Ethereum Virtual Machine (EVM)–compatible blockchain. The system models capital flows across four roles—Reserve Authority, National Banks, Local Banks, and Borrowers—using auditable smart contracts that enforce role-based permissions, reserve allocation constraints, and configurable loan repayment schedules. The design aims to preserve institutional hierarchy while enabling shared ledger visibility and automated rule execution.

To support risk management, the platform incorporates an off-chain analytics layer that applies machine learning techniques for transaction monitoring and decision support. A Random Forest classifier is used for supervised fraud detection, Isolation Forest for unsupervised anomaly detection, and SHAP-based feature attribution to provide interpretable risk signals for human reviewers. These tools are intended to augment, rather than replace, institutional credit governance.

The prototype is deployed on public EVM test networks, including Ethereum test environments and Polygon infrastructure, where observed confirmation times are on the order of seconds and transaction costs are negligible under test conditions. Performance evaluation focuses on functional validation of hierarchical controls, transparency properties, and operational workflows rather than production-scale financial performance.

We position blockchain in this context as a coordination technology that can provide shared state visibility, tamper-evident audit trails, and programmable process enforcement across multiple institutions. The paper further discusses governance design, security controls, regulatory considerations, and a phased deployment pathway involving academic validation and regulatory sandbox pilots.

By integrating hierarchical institutional structures with transparent ledger infrastructure and data-driven risk analytics, the Crypto World Bank explores a hybrid architecture for institutional decentralized finance that targets improved operational efficiency and expanded access to credit in emerging markets.


**Keywords:** Blockchain, Development Finance, Hierarchical Lending, Smart Contracts, DeFi, Financial Inclusion, Explainable AI, Governance

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# 1 Literature Review

## 1.1 Preliminaries

This section establishes key terminology and concepts used throughout the literature review and the project design.

**Decentralized Finance (DeFi).** DeFi refers to financial services built on blockchain platforms that operate without traditional intermediaries. Lending protocols enable users to borrow and lend assets through smart contracts, with interest rates typically determined algorithmically or by market utilization.

**Hierarchical lending.** In traditional development finance, capital flows through multiple tiers: supranational institutions (e.g., World Bank) allocate to national banks, which disburse to local banks, which ultimately reach end borrowers. This tiered structure enables risk distribution and local market access but introduces operational complexity.

**Smart contracts.** Self-executing programs deployed on a blockchain that execute when predefined conditions are met. In lending, smart contracts handle deposits, loan requests, approvals, repayments, and interest computation without human intermediaries.

**Over-collateralization.** A lending model where borrowers must pledge collateral worth more than the loan amount. This is the dominant model in DeFi (e.g., Aave, Compound) due to the absence of credit scoring; it excludes under-collateralized lending common in traditional finance.

**Explainable AI (XAI).** Techniques that make machine learning model predictions interpretable to humans. SHAP (SHapley Additive exPlanations) and LIME are prominent methods for attributing feature importance to individual predictions, critical for regulatory compliance in lending decisions.

**Proof of Stake (PoS).** A consensus mechanism where validators secure the network by staking tokens; used by Ethereum (post-Merge) and Polygon. PoS is more energy-efficient than Proof of Work and enables faster block finality.

## 1.2 Review of Existing Research

The design of the Crypto World Bank is informed by a comprehensive review of 16 peer-reviewed papers spanning seven domains.

**Decentralized Lending and DeFi Protocols.** Werner et al. [10] systematize DeFi protocol design in their SoK paper, identifying that existing lending platforms are uniformly pool-based and over-collateralized, lacking institutional hierarchy—a gap this project directly addresses. Bastankhah et al. [1] introduce an adaptive, data-driven DeFi lending protocol with a dual fast/slow control architecture, demonstrating that dynamic interest rate adjustment significantly outperforms static utilization curves used by Aave and Compound. An IEEE evaluation framework [9] provides standardized metrics for assessing protocol capital efficiency, risk management, and governance quality, revealing that higher capital efficiency correlates with higher risk exposure.

**ML Fraud Detection in Blockchain.** Palaiokrassas et al. [2] leverage ML for multichain DeFi fraud detection across 23 protocols (54M+ transactions), demonstrating that DeFi-specific behavioral features improve XGBoost and Neural Network classifiers to F1-scores of 0.76–0.85, compared to 0.08 with transactional features alone. Agarwal et al. [3] present RiskSEA, a scalable graph embedding system deployed at Coinbase that combines node2vec embeddings with behavioral features, achieving F1: 0.851 on all 266M Ethereum addresses. Rahouti et al. [16] survey ML approaches for Bitcoin security, finding Random Forest achieves 94%+ precision—supporting its selection as the primary fraud detection model in this system.

**Explainable AI in Lending.** Adom et al. [4] provide a direct comparison of LIME and SHAP on loan approval systems, finding SHAP provides deeper, more consistent feature attributions via Shapley values, while LIME offers faster runtime. Lundberg and Lee [11] introduce the SHAP framework itself, providing the theoretical foundation (local accuracy, missingness, consistency properties) that this system adopts for generating explainable loan risk assessments. Bracke et al. [17] apply SHAP to credit default models in a Bank of England regulatory context, demonstrating that SHAP feature attributions align with domain expert expectations and satisfy regulatory explainability requirements.

**Reinforcement Learning for Lending Policy.** Qu et al. [5] apply offline RL (CQL, BC, TD3-BC) to optimize Aave interest rates using historical data, with TD3-BC demonstrating superior performance during stress events (May 2021 crash, March 2023 USDC depeg). Kiatsupaibul et al. [6] formulate credit underwriting as an MDP, showing RL-based approaches discover superior underwriting policies compared to traditional greedy (static threshold) methods—validating our planned DQN/PPO modules for adaptive borrowing limits.

**CBDC and Financial Inclusion.** Tan [7] develops an IMF model showing CBDCs in developing countries can bank large unbanked populations through a two-tier distribution model (central bank → commercial banks → users) that directly parallels our four-tier hierarchy. Mhlanga [8] demonstrates through systematic review that blockchain technology facilitates financial inclusion across transactions, savings, credit provision, and insurance—supporting the Crypto World Bank's mission of transparent, programmable lending for underserved populations.

**Smart Contract Security and Governance.** Atzei et al. [13] catalog Ethereum smart contract attack vectors (reentrancy, integer overflow, access control), directly informing our use of OpenZeppelin's ReentrancyGuard and Solidity 0.8.20 overflow protection. Tolmach et al. [15] survey 42 formal verification tools, recommending multi-tool approaches combining static analysis (Slither) and symbolic execution (Mythril). Beck et al. [14] propose a three-dimensional blockchain governance framework (IT, network, business) that maps directly to our three-pillar governance structure. Liu et al. [12] introduce Isolation Forest for unsupervised anomaly detection—adopted as our secondary detection model for wallet behavior analysis where labeled fraud data is scarce.

**Identified Gaps.** Across these 16 papers, four gaps motivate the Crypto World Bank: (1) no hierarchical institutional model exists in DeFi lending [10, 1, 9]; (2) limited integration of AI/ML security analytics with DeFi lending [2, 3, 16]; (3) insufficient explainability in automated lending decisions [4, 11, 17]; and (4) inadequate governance frameworks for multi-tier decentralized finance systems [14]. Full detailed reviews (Abstract, Introduction, Motivation, Methodology, Results, Conclusion) for all 16 papers are available in the project documentation repository; Microsoft Word (.docx) versions: Literature Reviews.

### 1.3 Summary of Key Findings

The literature review yields the following consolidated findings that directly inform the Crypto World Bank design:

- **DeFi lending is structurally flat.** Existing protocols (Aave, Compound, MakerDAO) use pool-based or over-collateralized models. No protocol models institutional hierarchy analogous to development finance. CBDC research [7, 8] confirms that tiered distribution (central bank → commercial banks → users) is viable and supports financial inclusion.
- **ML improves fraud detection in blockchain contexts.** DeFi-specific behavioral features sub-

stantially outperform transactional features alone (F1: 0.76–0.85 vs. 0.08). Random Forest achieves 94%+ precision in Bitcoin security applications. Graph embeddings (RiskSEA) scale to hundreds of millions of addresses.

- **SHAP satisfies regulatory explainability.** Bank of England research [17] demonstrates that SHAP attributions align with domain expert expectations and meet prudential requirements. SHAP provides more consistent feature attributions than LIME for loan approval systems.
- **RL can optimize lending policy.** Offline RL outperforms static interest rate curves during stress events. Credit underwriting formulated as an MDP yields superior policies compared to greedy threshold methods.
- **Smart contract security requires layered defense.** Reentrancy, overflow, and access control vulnerabilities are well-documented. OpenZeppelin primitives and formal verification tools (Slither, Mythril) are recommended. Governance frameworks must address IT, network, and business dimensions.
- **Blockchain enables financial inclusion.** Systematic reviews confirm that blockchain facilitates transactions, savings, credit, and insurance for underserved populations. Transparent, programmable lending can reach the 1.4 billion unbanked adults globally.

These findings justify the Crypto World Bank's four-tier architecture, AI/ML integration (Random Forest, Isolation Forest, SHAP), and governance design.

# 2 Course Implementation

This project has been developed in conjunction with three undergraduate courses at BRAC University. The following course reports document the database design, software engineering methodology, and system analysis for the Crypto World Bank platform:

- **CSE370 – Database Management:** Relational schema design, ER/EER modeling, normalization, and indexing. View report
- **CSE470 – Software Engineering:** Agile methodology, sprint planning, user stories, and SDLC practices. View report
- **CSE471 – System Analysis and Design:** Use cases, sequence diagrams, activity diagrams, and component architecture. View report

*Microsoft Word (.docx) versions: All DOCX files.*

# 3 Problem Statement and Proposed Solution

## 3.1 Background

International development finance is delivered through layered institutional structures in which multilateral development institutions (e.g., World Bank, International Monetary Fund), national financial

intermediaries, and local lenders jointly channel capital to end borrowers. While this model enables risk distribution and local market access, it is associated with operational challenges: limited cross-tier transparency, lengthy cross-border settlement processes, high reconciliation and compliance overhead, and constrained scalability of risk assessment workflows.

Simultaneously, approximately 1.4 billion adults remain unbanked globally according to the World Bank Global Findex (2021), highlighting the need for more efficient and accessible credit delivery mechanisms. The Decentralized Finance (DeFi) ecosystem has demonstrated that smart contract platforms can automate lending with full transparency; however, existing DeFi protocols employ flat, pool-based architectures that do not model institutional hierarchy or integrate AI-driven risk analytics.

## 3.2 Rationale of the Study

This study is motivated by the convergence of three factors: (1) the persistent inefficiencies in traditional development finance that limit transparency and settlement speed; (2) the gap between DeFi's technical capabilities and its failure to address hierarchical, institutional lending use cases; and (3) the opportunity to combine blockchain's auditability with AI/ML security analytics to improve both operational efficiency and decision quality. By designing a prototype that bridges these domains, we aim to demonstrate a viable path toward transparent, programmable, and AI-augmented hierarchical lending for underserved populations.

## 3.3 Problem Statement

The global development finance ecosystem is characterized by a hierarchical topology in which supranational entities (e.g., the World Bank, International Monetary Fund) allocate capital to sovereign national institutions, which in turn disburse funds to regional and local banking entities, ultimately reaching end-user borrowers. This multi-layered intermediation introduces several systemic inefficiencies:

1. **Opacity of ledger state.** Internal bookkeeping across tiers is not publicly auditable. Stakeholders at lower tiers have limited visibility into reserve adequacy, approval criteria, and fund utilization at higher tiers.
2. **Settlement latency.** Cross-border capital transfers between tiers involve multiple intermediaries, compliance checkpoints, and reconciliation processes, resulting in settlement periods ranging from days to weeks.
3. **Manual risk assessment.** Creditworthiness evaluation and fraud detection are predominantly human-driven, introducing inconsistency, bias, and scalability constraints.
4. **Trust deficit.** The effort required to establish inter-institutional trust through legal agreements, audit engagements, and compliance certifications constitutes a significant overhead that disproportionately affects smaller institutions and borrowers in developing economies.

Simultaneously, the Decentralized Finance (DeFi) ecosystem has demonstrated that smart contract platforms can automate lending, collateral management, and interest computation with full transparency. However, existing DeFi protocols (e.g., Aave, Compound, MakerDAO) exhibit fundamental limitations in this context:

- They employ **peer-to-pool** or **over-collateralized** lending models that do not accommodate institutional hierarchies.
- They lack integration with **AI-driven risk analytics** and **explainable decision support** systems.
- They do not model **role-based, tiered governance** analogous to development finance institutions.

## 3.4 Objectives

The objectives of this project are:

1. To design and implement a four-tier hierarchical lending architecture (World Bank → National Bank → Local Bank → Borrower) on an EVM-compatible blockchain that preserves institutional hierarchy while enabling shared ledger visibility.
2. To integrate off-chain AI/ML security analytics—including fraud detection (Random Forest), anomaly identification (Isolation Forest), and explainable risk assessment (SHAP)—to augment human credit governance.
3. To demonstrate transparent, programmable lending workflows with configurable borrowing limits, installment schedules, and role-based access control enforced by smart contracts.
4. To validate the prototype on public testnets (Polygon, Ethereum Sepolia) with functional verification of hierarchical controls, transparency properties, and operational workflows.
5. To document governance design, security controls, regulatory considerations, and a phased deployment pathway for academic validation and potential regulatory sandbox pilots.

### 3.5 Methodology in Brief

The project adopts a lightweight Agile/Scrum methodology with three sprints over an 8-week development window. Sprint 1 establishes smart contracts, frontend scaffolding, wallet integration, and database schema. Sprint 2 delivers lending features (loan requests, approvals, installments, borrowing limits), chat system, income verification, and hierarchical bank registration. Sprint 3 integrates AI/ML fraud detection, risk dashboard, security audit, and documentation. Development uses Solidity 0.8.20 for smart contracts, React 18 with TypeScript for the frontend, FastAPI for the backend, and scikit-learn for ML models. The system is deployed on Polygon and Ethereum testnets with zero real-cryptocurrency cost.

### 3.6 Scopes and Challenges

**Scope.** The prototype operates exclusively on public testnets (no real cryptocurrency). In-scope features include: four-tier hierarchical lending, loan request and approval workflow, installment payments, borrowing limit computation, borrower–bank chat, income verification, fraud detection (Random Forest + SHAP), and anomaly detection (Isolation Forest). The dual-currency facility is designed as an auxiliary add-on to existing banking infrastructure. Out of scope for the prototype: mainnet deployment, fiat on-ramp/off-ramp integration, KYC/AML compliance automation, and production-scale stress testing.

**Challenges.** Key challenges include: (1) *Data scarcity*—labeled fraud data for DeFi lending is limited; synthetic or transfer learning may be required. (2) *Regulatory uncertainty*—crypto lending faces jurisdictional restrictions; the prototype mitigates by operating on testnets only. (3) *Gas cost sensitivity*—complex on-chain logic increases transaction cost; the design partitions AI/ML and heavy computation off-chain. (4) *Model interpretability*—regulatory requirements for explainable lending decisions are addressed via SHAP-based feature attribution.

### 3.7 Blockchain Justification

The problem described above is fundamentally a **multi-party coordination and trust challenge** involving disparate institutions with potentially misaligned incentives. We assert that blockchain technology addresses this class of problems more effectively than any conventional alternative for the following reasons:

1. **Trust minimization through consensus.** Distributed consensus mechanisms ensure that no single party can unilaterally alter the ledger state, eliminating the need for a trusted central operator.
2. **Programmable enforcement.** Smart contracts codify lending rules—borrowing limits, installment schedules, approval workflows—as deterministic, self-executing programs, reducing reliance on manual processes and bilateral agreements.

3. **Cryptographic auditability.** All state transitions are recorded in an immutable, publicly verifiable transaction log, enabling real-time auditing by regulators, partners, and borrowers without requiring costly third-party engagements.
4. **Composable incentive structures.** On-chain reputation systems, governance tokens, and programmable fee distributions can align incentives across all network participants.

A conventional cloud-hosted database infrastructure would necessitate a trusted central operator to maintain ledger integrity, enforce access control, and provide audit guarantees—reintroducing the very trust dependencies that the hierarchical lending model seeks to mitigate. Blockchain eliminates this single point of trust failure.

### 3.8 Proposed Solution

The Crypto World Bank implements a **four-tier, on-chain hierarchical lending architecture**:

- **Tier 1 — World Bank:** Maintains the global crypto reserve. Allocates capital to registered National Banks through smart contract-mediated lending operations.
- **Tier 2 — National Banks:** Borrow from the World Bank reserve. Lend to registered Local Banks within their jurisdiction. Aggregate risk exposure at the national level.
- **Tier 3 — Local Banks:** Borrow from National Banks. Process loan requests from individual borrowers. Employ designated approvers for loan lifecycle management.
- **Tier 4 — Borrowers:** Submit loan requests to Local Banks. Repay through configurable installment schedules. Accumulate on-chain repayment history that informs future borrowing limits.

The platform further integrates:

- **Risk-aware borrowing limits** computed from rolling 6-month and 1-year transaction windows.
- **Automatic installment generation** for loan amounts exceeding a configurable threshold (e.g., 100 ETH equivalent).
- **Off-chain AI/ML security analytics** for fraud detection (Random Forest), anomaly identification (Isolation Forest), and explainable risk assessment (SHAP).

## 4 Market Analysis and Partnership Ecosystem

### 4.1 Market Sizing

**Table 1:** Market segments.

| Segment | Description | Estimated Scale |
|---|---|---|
| Total Addressable Market (TAM) | Global decentralized lending market; DeFi lending TVL has historically exceeded $50B | $50B+ |
| Serviceable Addressable Market (SAM) | Institutional and semi-institutional lending requiring hierarchical structures | $5–10B |
| Serviceable Obtainable Market (SOM) | Pilot deployments in regulatory sandboxes, academic prototypes, NGO-backed microfinance | $50–200M |

***Sources:*** *(a) TAM: DefiLlama [19] tracks DeFi TVL; lending protocols have historically exceeded $50B in total value locked; (b) SAM and SOM: Project-derived estimates based on industry segmentation of institutional/semi-institutional lending and pilot/regulatory sandbox addressable market [20].*

The market for **transparent, hierarchy-preserving decentralized lending** is presently unaddressed by existing DeFi protocols, which universally adopt flat, pool-based architectures. This represents a significant whitespace opportunity.

## 4.2 Target Customer Segment

The Crypto World Bank targets the **retail customer segment** — individual borrowers and small businesses seeking transparent, accessible crypto-based lending services.

**Table 2:** Target customer segment profile.

| Characteristic | Description |
|---|---|
| Primary Users | Individual retail borrowers seeking personal or small business loans |
| Geographic Focus | Developing economies with limited traditional banking access (e.g., Bangladesh, Southeast Asia, Sub-Saharan Africa) |
| Loan Size Range | Micro to mid-range: 0.1 ETH – 500 ETH equivalent (~$200 – $1,000,000 at current rates) |
| User Profile | Digitally literate individuals with cryptocurrency wallet access; small business owners; gig-economy freelancers |
| Key Pain Points | High interest rates from informal lenders; lack of credit history in traditional systems; exclusion from banking due to documentation barriers |

**Why Retail:** The retail segment represents the largest underserved population in developing economies. According to the World Bank's Global Findex Database, approximately 1.4 billion adults remain unbanked, with the majority concentrated in developing countries. By targeting retail customers, the platform maximizes financial inclusion impact while generating the transaction volume necessary to sustain the lending ecosystem. The hierarchical banking model (World Bank → National → Local → Borrower) mirrors how traditional microfinance institutions reach retail customers in underserved regions, but with blockchain-enforced transparency and AI-enhanced risk assessment.

### 4.3 Partner Ecosystem

**Table 3:** Partner categories and roles.

| Partner Category | Functional Role | Blockchain-Mediated Incentive |
|---|---|---|
| Financial Regulators | Regulatory sandbox approval; compliance oversight | Reduced enforcement cost through on-chain transparency and audit trails |
| Banking Institutions | Network membership as National/Local Banks | Access to diversified global reserve; reduced inter-bank settlement friction |
| Payment Gateway Providers | Fiat-to-crypto on-ramp and off-ramp services | Volume-based transaction fees; expanded market reach |
| Academic & Research Institutions | Validation of AI/ML models; publication of research findings | Access to anonymized datasets; collaborative research opportunities |
| Non-Governmental Organizations | Pilot deployment; field testing with underserved borrower populations | Transparent, low-friction credit access for beneficiaries |

### 4.4 Incentive Alignment Through the Blockchain Platform

Partner incentives are allocated and enforced through the blockchain platform itself:

- **Immutable repayment records** serve as on-chain reputation signals, enabling data-driven credit decisions without third-party credit bureaus.
- **Transparent reserve verification** allows partners to independently confirm that allocated capital is deployed as committed.
- **Programmable fee structures** (future extension) distribute transaction fees proportionally to network participants based on their contribution to lending volume and governance.

## 5 Competitive Landscape and Risk Assessment

### 5.1 Direct Competition

**Table 4:** Direct competition.

| Competitor Category | Representative Examples | Crypto World Bank Differentiation |
|---|---|---|
| DeFi lending protocols | Aave, Compound, MakerDAO | Hierarchical four-tier institutional model; integrated AI/ML risk layer; role-based governance |
| Traditional development finance | World Bank, IFC, ADB | On-chain transparency; programmable enforcement; near-instant settlement |
| Blockchain-native credit platforms | Centrifuge, Maple Finance, Goldfinch | Combined hierarchy and AI security; designed for institutional adoption |

### 5.2 Indirect Competition

- **Central Bank Digital Currencies (CBDCs):** Emerging CBDC frameworks may address settlement efficiency but do not provide decentralized lending capabilities. The Crypto World Bank positions as a complementary **lending layer** operating on top of digital asset infrastructure.
- **Peer-to-peer lending platforms:** Predominantly fiat-denominated and centralized; the Crypto World Bank targets crypto-native and hybrid institutional use cases.

### 5.3 Risk Taxonomy and Mitigation Strategy

**Table 5:** Risk taxonomy and mitigation.

| Risk Category | Description | Severity | Mitigation |
|---|---|---|---|
| Partner non-cooperation | Key ecosystem partners decline to participate | Medium | Initiate with low-barrier academic and NGO pilots; minimize switching costs |
| Incentive misalignment | Participant incentives diverge from network objectives | Medium | Codify rules in smart contracts; implement governance token alignment (future) |
| Smart contract vulnerability | Exploit in contract logic (reentrancy, overflow) | High | OpenZeppelin security primitives; formal audit (planned); pause mechanism |
| Regulatory adversity | Jurisdictional restrictions on crypto lending | Medium | Operate exclusively on testnets during prototype phase; engage regulatory sandbox programs |
| Market adoption failure | Insufficient user or institutional traction | Low | Leverage academic and competition channels; open-source codebase |
| AI/ML model degradation | Fraud detection accuracy decays over time | Low | Continuous model retraining pipeline; human-in-the-loop review; explainability via SHAP |

### 5.4 Comparative Advantage

- **Value delivery superiority:** Blockchain-enforced transparency eliminates reconciliation overhead; AI/ML augments human judgment with data-driven risk scores.
- **Performance superiority:** Sub-minute transaction confirmation on EVM Layer-2 networks (Polygon) versus multi-day settlement in traditional development finance.

## 6 System Architecture and Governance Framework

### 6.1 High-Level Architecture

The system employs a **three-layer decentralized application architecture**:

**PRESENTATION LAYER**

**React 18** · **TypeScript** · **Material Design 3** · **Wagmi** · **Viem**

Modules: Dashboard, Deposit, Loan, Admin, Risk AI, QR

↓

**SMART CONTRACT LAYER** (EVM: Ethereum / Polygon)

**World Bank Reserve Contract** · **National Bank Contract** · **Local Bank Contract**

Operations: Reserve Mgmt, Hierarchical Lending, Loan Lifecycle, Access Control, Emergency Controls

↓

**OFF-CHAIN SERVICES LAYER**

**PostgreSQL** (Relational DB) · **FastAPI** (REST API)

AI/ML: Random Forest (Fraud) · Isolation Forest (Anomaly) · SHAP (Explainability)

Event Listener · Cache (Redis)

## 6.2 Blockchain Platform Selection

**Table 6:** Blockchain platform selection.

| Criterion | Selection | Justification |
|---|---|---|
| Platform | Ethereum Virtual Machine (EVM) | Largest developer ecosystem; battle-tested security model; extensive tooling |
| Network | Polygon Mumbai / Ethereum Sepolia (testnets) | Zero-cost deployment; production-equivalent behavior; free faucet access |
| Consensus | Proof-of-Stake (PoS) via Polygon validators | Energy-efficient; sub-2-second block finality; decentralized validator set |
| Smart contract language | Solidity 0.8.20 | Industry standard; mature compiler with overflow protection; rich library ecosystem |

## 6.3 Transaction Verification and Consensus

- **Polygon PoS:** Transactions are validated by a decentralized set of PoS validators. Block finality is achieved within approximately 2 seconds. Checkpoints are periodically committed to Ethereum mainnet for additional security.
- **On prototype testnets:** Mumbai and Sepolia employ analogous consensus models with no financial cost, enabling iterative development and testing.

## 6.4 On-Chain and Off-Chain Data Partitioning

**Table 7:** Data partitioning.

| Data Category | Storage | Rationale |
|---|---|---|
| Reserve balances, loan requests, approval/rejection events, repayment transactions | On-chain | Immutability, public auditability, trustless verification |
| User profiles, income verification documents, chat messages, AI/ML inference logs | Off-chain (PostgreSQL) | Data privacy, query flexibility, storage cost optimization |
| Borrowing limit computations | Off-chain with on-chain enforcement | Complex temporal aggregation; results committed as on-chain constraints |
| Cryptocurrency market data | Off-chain (cached) | High-frequency updates; external API dependency |

## 6.5 Data Model

The relational database schema comprises 15 normalized entities in Third Normal Form (3NF):

**Core hierarchy:** `WORLD_BANK`, `NATIONAL_BANK`, `LOCAL_BANK`, `BANK_USER`, `BORROWER`

**Lending lifecycle:** `LOAN_REQUEST`, `INSTALLMENT`, `TRANSACTION`, `BORROWING_LIMIT`

**Communication:** `CHAT_MESSAGE`

**AI/ML and security:** `AI_CHATBOT_LOG`, `AI_ML_SECURITY_LOG`

**Supporting:** `INCOME_PROOF`, `MARKET_DATA`, `PROFILE_SETTINGS`

A comprehensive Entity-Relationship Diagram (ERD) is maintained in the project documentation repository.

## 6.6 Digital Identity System

- **Wallet-based identity:** Users authenticate via Ethereum wallet signatures (MetaMask, WalletConnect). No centralized credential store is required.
- **Role binding:** Wallet addresses are mapped to hierarchical roles (Owner, National Bank, Local Bank, Approver, Borrower) within the smart contract permission system.

## 6.7 Integration with Legacy Systems

- **Current:** Standard Web3 wallet integration (MetaMask browser extension, WalletConnect mobile protocol).
- **Planned:** Fiat on-ramp/off-ramp via payment gateway APIs; KYC/AML compliance via external identity verification providers.

## 6.8 Auxiliary Dual-Currency Facility

The Crypto World Bank's cryptocurrency exchange capability is designed as an **auxiliary facility** integrated into existing banking infrastructure worldwide. Rather than operating as a standalone

exchange, the platform extends the services of participating banks by enabling a **dual-currency feature**—allowing eligible bank customers to transact in both fiat and cryptocurrency within their existing banking relationship.

- **Integration model:** The dual-currency facility is offered by all participating banks (World Bank, National Banks, and Local Banks) as an add-on service to their existing product portfolio. No new banking license or standalone entity is required.
- **Eligibility determination:** Eligibility for dual-currency services is determined by the bank officers who manage lending operations. These officers assess customer suitability based on existing KYC/AML compliance, account standing, and lending relationship history.
- **No defaulting of conditions:** The project does not override, modify, or default any existing banking conditions, regulatory requirements, or contractual obligations. All existing banking terms, interest rate agreements, and compliance frameworks remain fully in effect.
- **Scope:** The facility enables fiat-to-crypto and crypto-to-fiat conversions, cryptocurrency-denominated lending and repayment, and transparent on-chain transaction records—all within the governance structure of the participating bank.

This design ensures that the Crypto World Bank **complements rather than disrupts** existing financial infrastructure, reducing adoption barriers for traditional banking institutions and their customers.

## 6.9 Regulatory Compliance Considerations

As the platform operates within the regulated banking domain:

- The prototype phase operates exclusively on **public testnets** with no real monetary value.
- The architecture supports **audit log generation** for regulatory review.
- Future production deployment would engage **regulatory sandbox programs** in target jurisdictions.

## 6.10 Governance Framework

### 6.10.1 Network Membership Governance

**Table 8:** Network membership governance.

| Governance Aspect | Implementation |
|---|---|
| Member on-boarding | World Bank owner registers National Banks; National Banks register Local Banks; Local Banks designate approvers — all enforced on-chain |
| Member off-boarding | Deactivation flags in smart contracts; cascading access revocation |
| Regulatory oversight | Audit log emission via smart contract events; planned read-only regulator dashboard |
| Permission structure | Hierarchical: Owner → National Bank → Local Bank → Approver → Borrower; enforced by on-chain role-check modifiers |
| Network operations | Pause/unpause mechanism for emergency response; emergency withdrawal for critical situations |

### 6.10.2 Business Network Governance

**Table 9:** Business network governance.

| Governance Aspect | Implementation |
|---|---|
| Business charter | Defined in this whitepaper; operational parameters codified in smart contract constants |
| Common services | Reserve management, loan lifecycle orchestration, event-driven notification system |
| Business SLA | Testnet phase: best-effort availability. Production phase: 99.5% target uptime with multi-region deployment |
| Regulatory compliance | Architecture designed for audit trail generation; data partitioning supports GDPR-style data subject requests |

### 6.10.3 Technology Infrastructure Governance

**Table 10:** Technology infrastructure governance.

| Governance Aspect | Implementation |
|---|---|
| Distributed IT structure | Client-side frontend (decentralized delivery); blockchain layer (fully decentralized); backend API (centralized, horizontally scalable) |
| Technology assessment | Continuous evaluation of EVM alternatives (L2 rollups, sidechains) for cost and performance optimization |
| On-chain / off-chain data services | Clearly partitioned (see Section IV.D); event listeners synchronize state between layers |
| Risk mitigation | Smart contract pause mechanism; ReentrancyGuard; input validation; planned formal security audit |

## 6.11 Asset Tokenization

- **Current implementation:** Native blockchain currency (ETH/MATIC) serves as the reserve and loan denomination.
- **Planned extension:** ERC-20 stablecoin support (USDC, USDT) for USD-denominated lending operations; tokenized collateral instruments for under-collateralized lending scenarios.

## 6.12 Transaction Economics: Gas Fees and Interest Rates

### 6.12.1 Gas Fee Policy

All blockchain transaction costs (gas fees) are borne by the **transaction initiator**, following standard Ethereum/Polygon network conventions:

**Table 11:** Gas fee responsibility by transaction type.

| Transaction Type | Gas Fee Payer | Rationale |
|---|---|---|
| Loan Request | Borrower | Borrower initiates the loan request transaction |
| Loan Approval | Bank Approver | Approver initiates the approval transaction |
| Loan Rejection | Bank Approver | Approver initiates the rejection transaction |
| Installment Payment | Borrower | Borrower initiates each installment payment |
| Deposit to Reserve | Depositor | Depositor initiates the reserve contribution |
| Bank Registration | Registering Authority | World Bank Admin or National Bank initiates registration |

On **Polygon PoS** (the primary target network), gas fees are denominated in MATIC and are significantly lower than Ethereum mainnet — typically **$0.001–$0.01 per transaction**, making the platform economically viable for retail users with smaller loan amounts. This low fee structure is critical for the retail customer segment, where even small transaction costs can disproportionately affect borrower economics.

### 6.12.2 Interest Rate Structure

The platform implements **market-standard interest rates** aligned with prevailing DeFi lending benchmarks:

**Table 12:** Interest rate parameters.

| Parameter | Value | Benchmark |
|---|---|---|
| Base Annual Interest Rate | 5–12% APR | Aligned with Aave/Compound variable rates for major crypto assets |
| Rate Determination | Set by Local Bank approvers within World Bank-defined bounds | Configurable per-bank to reflect local market conditions |
| Late Payment Penalty | 2% of installment amount + 0.5% per additional week overdue (capped at 10%) | Industry-standard late fee structure |
| Interest Calculation | Simple interest on outstanding principal | Transparent, borrower-friendly method |
| Rate Transparency | All rate parameters stored on-chain as smart contract constants | Publicly auditable; no hidden fees |

***Sources:*** *Base rate range aligned with Aave and Compound variable rates for major crypto assets; see Aave docs (https://docs.aave.com/) and Compound docs (https://docs.compound.finance/). Late fee structure follows industry conventions for installment lending.*

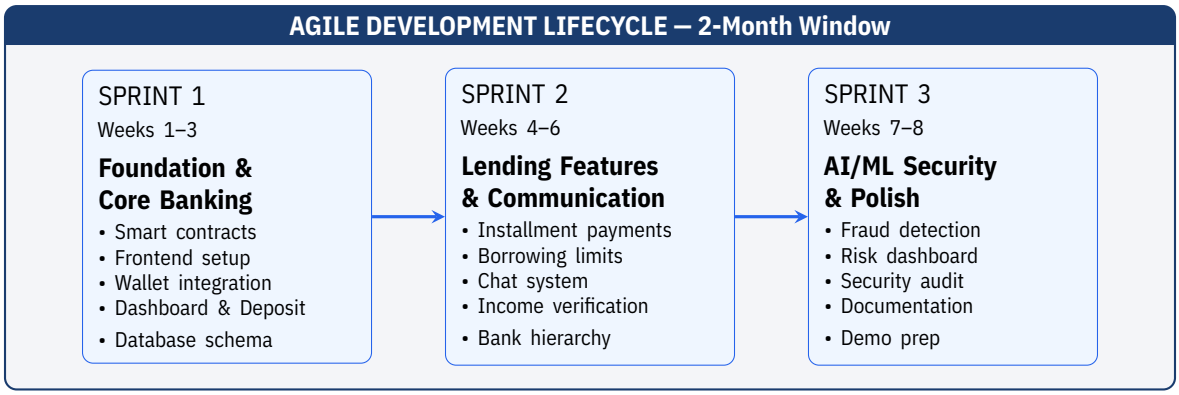
Interest rates are designed to balance lending sustainability with borrower affordability. The on-chain storage of rate parameters ensures borrowers can independently verify loan terms before execution. Local Banks may adjust rates within the bounds set by the World Bank tier, allowing localized pricing that reflects regional economic conditions.

# 7 Methodology

This paper and project planning have been structured in accordance with the Blockchain Olympiad Bangladesh AI **guidelines** and the CSE final-year project **rubrics**.

## 7.1 Development Methodology

The project adopts a **lightweight Agile/Scrum** methodology tailored for an academic prototype with a fixed two-month development window. The full methodology and sprint backlog are documented in the CSE470 Software Engineering document. The iterative approach enables incremental delivery of demonstrable features while accommodating evolving requirements inherent in research-oriented development.



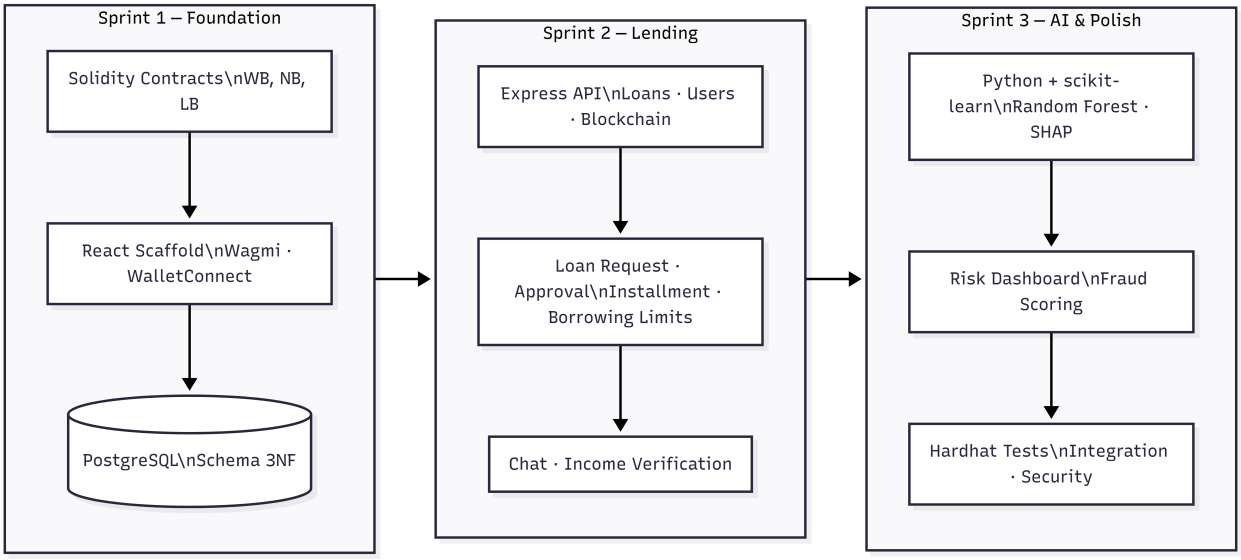




**Figure 1:** Technical development methodology: blockchain (Solidity), frontend (React, Wagmi, WalletConnect), backend (Express API, PostgreSQL), and AI/ML (Python, scikit-learn, Random Forest, SHAP) components across the three sprints.

The complete project source code, smart contracts, frontend, backend, and documentation are available in the main repository (Microsoft Word report versions).

## 7.2 Agile Sprint Plan (8-Week Schedule)

**Table 13:** Sprint plan.

| Sprint | Duration | Goal | Key Deliverables |
|---|---|---|---|
| Sprint 1 | Weeks 1–3 | Foundation and Core Banking | Smart contract suite (WorldBankReserve, NationalBank, LocalBank); React frontend scaffolding; wallet integration; Dashboard and Deposit pages; database schema (15 tables, 3NF) |
| Sprint 2 | Weeks 4–6 | Lending Features and Communication | Loan request and approval workflow; installment payment system; borrowing limit engine; borrower–bank chat; income verification; hierarchical bank registration |
| Sprint 3 | Weeks 7–8 | AI/ML Security and Finalization | Fraud detection model (Random Forest + SHAP); risk dashboard; anomaly detection prototype; security audit; documentation; demo preparation |

## 7.3 Sprint Deliverables Summary

**Sprint 1 — Foundation:** Smart contracts (WorldBankReserve, NationalBank, LocalBank), wallet integration, dashboard, database schema (15 tables, 3NF).

**Sprint 2 — Lending:** Loan request/approval workflow, installment payments, borrowing limits, borrower–bank chat, income verification, hierarchical bank registration.

**Sprint 3 — AI/ML and Polish:** Fraud detection (Random Forest + SHAP), risk dashboard, anomaly detection, security audit, documentation. The full user story backlog (US-1.1 through US-3.12) is in the CSE470 Software Engineering document.

## 7.4 SDLC Stage Mapping

The project maps to the seven stages of the Software Development Life Cycle (SDLC):

**Table 14:** SDLC stage mapping.

| SDLC Stage | Project Activity | Deliverable |
|---|---|---|
| 1. Planning & Requirements | Feasibility studies; professor consultations; BCOLBD guideline review | Feasibility Report; Project Plan |
| 2. Defining Requirements | System analysis (CSE471); use case definitions; constraints | CSE471 document; UC-1 through UC-5 |
| 3. Designing Architecture | Three-layer architecture; DB schema (3NF); smart contract interfaces | Architecture diagrams; ERD; DFD |
| 4. Development | Sprint 1–3: smart contracts, frontend, backend, AI/ML | Source code; DApp prototype |
| 5. Testing | Hardhat unit tests (12+); integration testing; AI/ML evaluation | Test reports; model metrics |
| 6. Deployment | Testnet deployment; frontend (Vercel); backend (Render) | Live prototype on testnet |
| 7. Maintenance | Monitoring; model retraining; bug fixes; iteration | Updated docs; retrained models |

### 7.5 Design Decisions and Alternatives Considered

For each major design decision, alternatives were evaluated and the first and second choices justified. The complete analysis is documented in the project decision justification report.

**Table 15:** Summary of design decisions.

| Decision Area | 1st Choice | 2nd Choice | Key Criterion |
|---|---|---|---|
| Methodology | Agile / Scrum | Incremental | Evolving scope, milestones |
| Architecture | DApp + Off-chain AI | Hybrid w/ Oracle | Gas cost, ML flexibility |
| Frontend | React + TypeScript | Vue + TypeScript | Web3 ecosystem |
| Smart Contract | EVM (Solidity) | Solana (Rust) | Ecosystem, free testnets |
| UI Design | MUI (Material 3) | Tailwind CSS | Banking UI, speed |
| Fraud Detection | Random Forest | XGBoost | SHAP, 94%+ precision |
| Anomaly Detection | Isolation Forest | Autoencoder | Unsupervised, no labels |
| XAI Method | SHAP | LIME | Theory, regulatory |
| Database | PostgreSQL | SQLite | CSE370, 3NF, async |
| Hosting | Vercel + Render | Localhost | $0 cost, public URL |

## 8 Feasibility Analysis

## 8.1 Technical Feasibility

**Table 16:** Technical feasibility.

| Component | Assessment | Evidence |
|---|---|---|
| Smart contracts | **Fully feasible** | Three contracts implemented, compiled, and tested with Hardhat (12+ passing unit tests); Solidity 0.8.20 with OpenZeppelin |
| Frontend DApp | **Fully feasible** | React 18 + TypeScript application with all pages implemented; Wagmi and RainbowKit provide mature wallet integration |
| Blockchain deployment | **Fully feasible** | Polygon Mumbai and Ethereum Sepolia testnets provide zero-cost, production-equivalent environments with free faucet access |
| AI/ML integration | **Feasible w/ constraints** | Random Forest inference on tabular data achieves sub-50ms latency; SHAP explanations computable in real-time; prototype scope limits to fraud detection |
| Database backend | **Feasible** | PostgreSQL schema designed (15 tables, 3NF); FastAPI provides async REST framework with automatic OpenAPI documentation |

**Conclusion:** All core components are technically feasible within the prototype scope. The smart contract and frontend layers are already implemented and validated.

## 8.2 Economic Feasibility

**Table 17:** Economic feasibility.

| Cost Category | Estimate | Notes |
|---|---|---|
| Blockchain deployment | $0 | Public testnets (Sepolia, Mumbai) — no real cryptocurrency required |
| Frontend hosting | $0 | Vercel free tier or localhost for demo |
| Backend hosting | $0 | Render free tier or localhost |
| AI/ML training | $0 | Local machine (16 GB RAM, 16 GB VRAM) or Google Colab free tier |
| Development tools | $0 | Hardhat, VS Code, Git — all open-source |
| **Total prototype cost** | **$0** | Entire prototype operates at zero financial cost |

**Conclusion:** The prototype is economically viable at zero cost, leveraging free-tier services and open-source tooling.

## 8.3 Operational and Schedule Feasibility

**Table 18:** Operational phases.

| Phase | Timeline / Activities |
|---|---|
| Pre-Thesis 1 | Completed: Requirements analysis, system modeling, database design, software engineering planning, initial prototype |
| Pre-Thesis 2 | Current (8 weeks): Smart contract finalization, frontend completion, fraud detection implementation, documentation |
| Final Thesis | 4–8 weeks: Additional ML experiments, evaluation, final report, competition submission |

The 8-week Agile plan distributes work across three sprints with defined deliverables per sprint. Risk mitigation includes: **Buffer:** Sprint 3 includes documentation and polish time as contingency. **Scope flexibility:** AI/ML features beyond fraud detection are designated as exploratory/future work. **Parallel workstreams:** Frontend, smart contract, and documentation tasks can proceed concurrently.

## 8.4 Revenue Projection and Economic Impact

The following projection models the annual revenue potential of the hierarchical lending system at full deployment scale. Calculations use a reference ETH price of **$2,000** (February 2026 market rate) and interest rate parameters defined in Section IV.I.

***Sources for revenue assumptions:*** *ETH price from spot market data (e.g., CoinGecko, https://www.coingecko.com/); interest rate tiers (3%, 5%, 8% APR) aligned with wholesale inter-bank and retail DeFi benchmarks; default rate (3%) and origination fee (0.25%) from conservative DeFi industry estimates.*

**Table 19:** Revenue projection assumptions.

| Parameter | Value |
|---|---|
| Reference ETH price | $2,000 (Feb 2026) |
| World Bank → National Bank rate | 3% APR (wholesale inter-bank) |
| National Bank → Local Bank rate | 5% APR (inter-bank) |
| Local Bank → Borrower rate | 8% APR (retail lending, mid-range of 5–12%) |
| Average loan term | 12 months |
| Loan turnover factor | 2× per year (6-month avg effective cycle) |
| Default rate provision | 3% (conservative DeFi industry estimate) |
| Origination fee | 0.25% per disbursement |

### 8.4.1 Tier 1 — World Bank (Global Reserve)

**Table 20:** World Bank revenue projection (1,000,000 ETH reserve).

| Metric | ETH | USD Equivalent |
|---|---|---|
| Total reserve capital | 1,000,000 | $2,000,000,000 |
| Disbursed to 5 National Banks | 1,000,000 | $2,000,000,000 |
| Annual interest income (3%) | 30,000 | $60,000,000 |
| Origination fees (0.25%) | 2,500 | $5,000,000 |
| Less: Default provision (3%) | −975 | −$1,950,000 |
| **Net annual revenue** | **31,525** | **$63,050,000** |

### 8.4.2 Tier 2 — National Banks (5 Banks × 200,000 ETH)

**Table 21:** National Bank revenue projection (per bank and combined).

| Metric (per National Bank) | ETH | USD Equivalent |
|---|---|---|
| Capital received from World Bank | 200,000 | $400,000,000 |
| Cost of capital (3% to World Bank) | −6,000 | −$12,000,000 |
| Disbursed to 10 Local Banks (10,000 each) | 100,000 | $200,000,000 |
| Interest income from Local Banks (5%) | 5,000 | $10,000,000 |
| Retained reserve direct lending (100,000 ETH at 6%) | 6,000 | $12,000,000 |
| Origination fees (0.25%) | 500 | $1,000,000 |
| Less: Default provision (3%) | −345 | −$690,000 |
| **Net annual revenue per bank** | **5,155** | **$10,310,000** |
| **Combined (5 banks)** | **25,775** | **$51,550,000** |

### 8.4.3 Tier 3 — Local Banks (50 Banks × 10,000 ETH)

**Table 22:** Local Bank revenue projection (per bank and combined).

| Metric (per Local Bank) | ETH | USD Equivalent |
|---|---|---|
| Capital received from National Bank | 10,000 | $20,000,000 |
| Cost of capital (5% to National Bank) | −500 | −$1,000,000 |
| Loans issued: 20 borrowers × 1,000 ETH (2× turnover) | 20,000 | $40,000,000 |
| Interest income from borrowers (8%) | 1,600 | $3,200,000 |
| Origination fees (0.25% on 20,000 ETH) | 50 | $100,000 |
| Less: Default provision (3%) | −49.5 | −$99,000 |
| **Net annual revenue per bank** | **1,100.5** | **$2,201,000** |
| **Combined (50 banks)** | **55,025** | **$110,050,000** |

### 8.4.4 Borrower Economics (1,000 Borrowers × 1,000 ETH)

**Table 23:** Borrower-level economics.

| Metric (per Borrower) | ETH | USD Equivalent |
|---|---|---|
| Average loan amount | 1,000 | $2,000,000 |
| Annual interest cost (8%) | 80 | $160,000 |
| Estimated business return (15–20% avg.) | 150–200 | $300,000–400,000 |
| **Net borrower surplus (after interest)** | **70–120** | **$140,000–240,000** |
| **Aggregate (1,000 borrowers)** | **70,000–120,000** | **$140M–240M** |

### 8.4.5 System-Wide Revenue Summary

**Table 24:** System-wide annual revenue summary.

| Tier | Annual Revenue (ETH) | USD Equivalent |
|---|---|---|
| Tier 1: World Bank (1 entity) | 31,525 | $63,050,000 |
| Tier 2: National Banks (5 entities) | 25,775 | $51,550,000 |
| Tier 3: Local Banks (50 entities) | 55,025 | $110,050,000 |
| **Total platform revenue** | **112,325** | **$224,650,000** |
| Borrower surplus generated | 70,000–120,000 | $140M–240M |

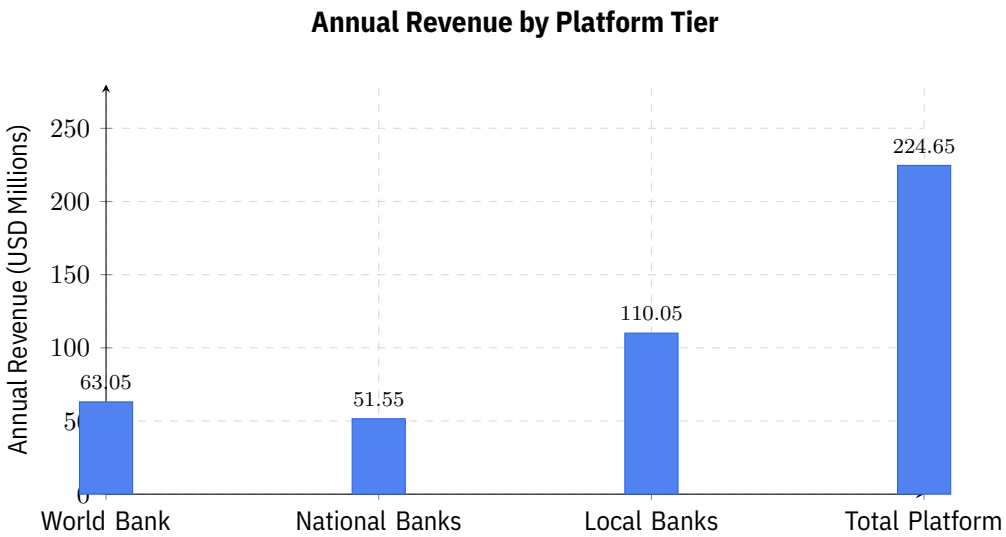



**Figure 2:** Annual revenue projection by tier (USD millions, at $2,000/ETH).

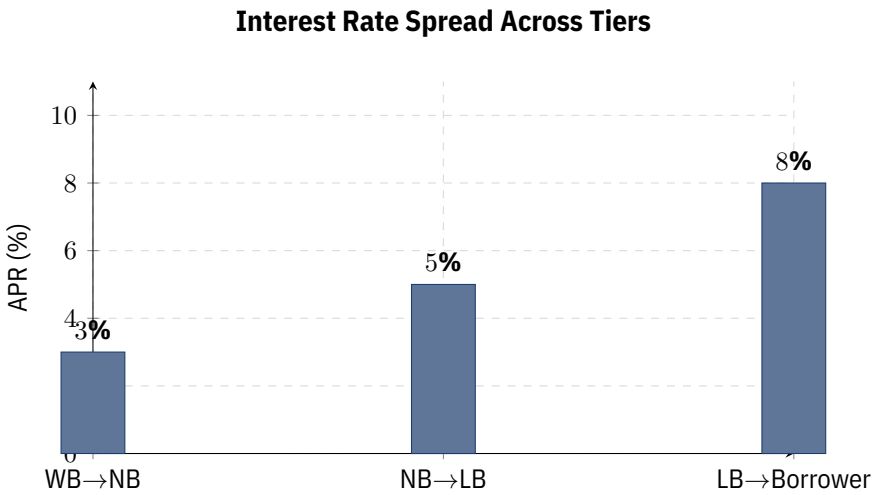



**Figure 3:** Hierarchical interest rate spread (APR) across the four-tier lending structure.

### 8.4.6 Global Economic Impact

Beyond direct platform revenue, the Crypto World Bank creates broader economic value:

- **Capital deployment:** $2 billion in lending capital reaches 1,000 direct borrowers across developing economies. According to World Bank research on development finance, each dollar of institutional lending generates approximately $2.5–3.0 in downstream economic activity through the fiscal multiplier effect, implying a potential **$5–6 billion annual economic stimulus**.
- **Job creation:** SME lending research (IFC, 2019) estimates that every $50,000 in small business credit creates or sustains one job. At $2 billion in deployed capital, the platform could support an estimated **40,000 direct jobs** in borrower enterprises.
- **Financial inclusion:** Approximately 1.4 billion adults remain unbanked globally (World Bank Global Findex, 2021). The platform's retail-focused design provides transparent, accessible credit to underserved populations without requiring traditional banking relationships or credit histories.
- **Transaction cost reduction:** DeFi lending eliminates intermediary costs associated with traditional cross-border development finance. Settlement occurs in minutes rather than days, and on Polygon PoS, transaction fees remain below $0.01—representing a **60–80% cost reduction** compared to traditional correspondent banking.
- **Transparency dividend:** On-chain auditability reduces compliance and audit costs for participating institutions. Traditional bank audits cost $500,000–$2 million annually; blockchain-native audit trails significantly reduce this overhead.

***Sources for economic impact estimates:*** *(a) GDP multiplier ($2.5–3.0×): World Bank, "World Development Report 2022: Finance for an Equitable Recovery," https://www.worldbank.org/en/publication/wdr2022; (b) Job creation ($50K per job): International Finance Corporation (IFC), "MSME Finance Gap 2019," https://www.ifc.org/en/insights-reports/2019/msme-finance-gap; (c) 1.4 billion unbanked: World Bank, "The Global Findex Database 2021," https://www.worldbank.org/en/publication/globalfindex; (d) DeFi cost reduction (60–80%): Bank for International Settlements, "DeFi lending: intermediation without information?," BIS Working Paper No. 1183, 2024, https://www.bis.org/publ/work1183.pdf; (e) Bank audit costs ($500K–$2M): Deloitte, "Global Banking Outlook 2024."*

# 9 System Modeling

### 9.1 Use Case Diagram

**Actors (4):** Borrower, Bank Approver, World Bank Admin, National Bank — **29 use cases total.** A high-level use case diagram is maintained in the project documentation repository.

### 9.2 Sequence Diagram

The loan request, AI risk assessment, and approval flow span **49 interaction steps** across 9 participants. Full sequence diagrams are maintained in the project documentation repository.

### 9.3 Entity-Relationship Diagram

The ERD comprises 15 entities in 3NF. Key relationships:

- `WORLD_BANK` (1) → (N) `NATIONAL_BANK` → (N) `LOCAL_BANK` — Hierarchical banking structure
- `BORROWER` (1) → (N) `LOAN_REQUEST`; `LOAN_REQUEST` (1) → (N) `INSTALLMENT`, `CHAT_MESSAGE`; `BORROWER` (1) → (1) `BORROWING_LIMIT`
- `LOAN_REQUEST` / `TRANSACTION` → (N) `AI_ML_SECURITY_LOG`

Full ERD is in the project documentation repository.

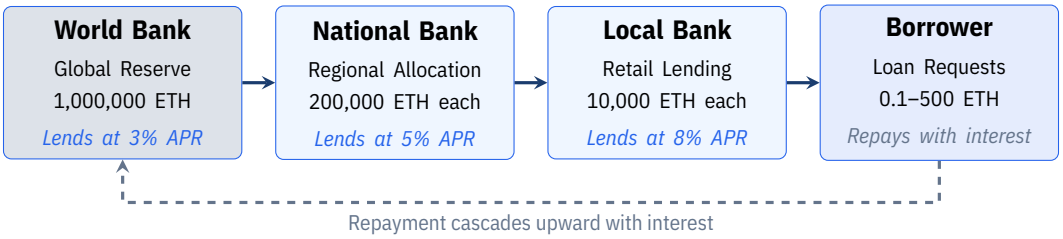



**Figure 4:** Four-tier hierarchical capital flow with cascading repayment.

### 9.4 Activity, Component, and Data Flow Diagrams

Activity diagrams (loan request to repayment, income verification, chat system, AI chatbot, hierarchical banking, market data, profile management), component diagrams (presentation, smart contract, backend, external subsystems), and data flow diagrams (Level 0 context and Level 1 with 12 processes and 11 data stores) are documented in the project documentation repository (DOCX versions).

## 10 Valuation and Distribution Strategy

## 10.1 Value Proposition

**Table 25:** Value proposition.

| Value Type | Description | Beneficiary |
|---|---|---|
| Operational efficiency | Elimination of manual reconciliation and multi-day settlement | Banking institutions |
| Audit cost reduction | Real-time on-chain auditability replaces periodic third-party audits | Regulators, institutions |
| Risk mitigation | AI/ML-augmented loan decisioning reduces default rates | Lenders, borrowers |
| Financial inclusion | Transparent, programmable lending accessible to underserved populations | Borrowers in developing economies |
| Research contribution | Novel intersection of hierarchical DeFi, AI security, and governance | Academic community |

## 10.2 Revenue Model

**Table 26:** Revenue model.

| Revenue Stream | Mechanism | Phase |
|---|---|---|
| Loan origination fees | Configurable percentage (0.1–0.5%) deducted at disbursement via smart contract | Phase 2 |
| Premium analytics services | Subscription-based access to advanced AI risk dashboards for institutional users | Phase 3 |
| Governance token economics | Network participation incentives distributed via governance token | Phase 3 |
| Grant and pilot funding | Academic research grants; regulatory sandbox program funding | Phase 1–2 |

## 10.3 Go-to-Market Strategy

**Table 27:** Go-to-market phases.

| Phase | Activities | Timeline |
|---|---|---|
| Phase 1: Academic Validation | Competition submission (BCOLBD 2025); thesis publication; open-source release | Current |
| Phase 2: Pilot Deployment | Regulatory sandbox application; partnership with one institutional entity; testnet-to-mainnet migration | 6–12 months |
| Phase 3: Production Scale | Multi-chain deployment; full AI/ML backend; commercial partnerships; governance token launch | 12–24 months |

### 10.4 Launch Readiness

- Whitepaper and technical documentation: **Published.**
- Smart contract suite: **Implemented and tested** (WorldBankReserve, NationalBank, LocalBank).
- Frontend DApp: **Implemented** (all pages, responsive, wallet integration).
- Prototype deployment: **Ready** for testnet deployment.
- AI/ML backend: **Architecture defined**; implementation in progress.

## 11 Academic Course Requirements Fulfillment

The Crypto World Bank project integrates deliverables that satisfy the requirements of multiple undergraduate courses in the B.Sc. Computer Science program at BRAC University. The following summarizes how each course's objectives have been addressed within this project.

### 11.1 CSE370 — Database Management

The project fulfills database management requirements through the design and implementation of a comprehensive relational data model. Deliverables include:

- A fully normalized Entity-Relationship (ER) diagram modeling the four-tier institutional hierarchy (World Bank, National Banks, Local Banks, Borrowers), loan lifecycle states, and transaction records.
- Relational schema design with referential integrity constraints, primary and foreign key relationships, and indexing strategies for efficient query execution.
- A PostgreSQL database implementation supporting off-chain storage of user profiles, transaction metadata, audit logs, and AI/ML feature tables.
- Database-level access control and role-based permission modeling aligned with the on-chain role hierarchy.

### 11.2 CSE470 — Software Engineering

The project addresses software engineering principles through systematic planning, modeling, and development practices. Deliverables include:

- Requirements analysis documented through functional and non-functional requirement specifications.
- Use case diagrams describing interactions among all four tiers of users and the system.
- Sequence diagrams illustrating end-to-end workflows for deposit, loan request, approval, repayment, and fraud detection processes.
- Activity diagrams capturing the control flow of key operations including hierarchical fund distribution and cascading repayment.
- Component diagrams depicting the system's modular architecture across the presentation, smart contract, and off-chain service layers.
- Data flow diagrams (Level 0 and Level 1) mapping information movement between system components.
- An Agile development methodology with a three-sprint, eight-week schedule, sprint backlogs, and burndown tracking.
- SDLC stage mapping, design decision analysis, Gantt charts, and risk assessment matrices.
- Revenue projection bar charts and comparative analysis visualizations.

### 11.3 CSE471 — System Analysis and Design

The project satisfies system analysis requirements through structured analytical modeling of the platform. Deliverables include:

- Context-level and decomposition data flow diagrams analyzing information flow across the four-tier hierarchy.
- Activity diagrams for each major system process: user registration, deposit flow, loan origination, installment repayment, fund distribution, and fraud alert handling.
- Component interaction diagrams showing dependencies between the React frontend, smart contract layer, REST API, PostgreSQL database, and AI/ML analytics engine.
- Sequence diagrams modeling synchronous and asynchronous interactions including wallet connection, blockchain transaction submission, event listening, and off-chain data synchronization.
- Feasibility analysis covering technical, economic, and operational dimensions.

### 11.4 CSE446 — Blockchain and Cryptocurrencies

The project directly applies blockchain and cryptocurrency concepts through the development of production-grade Solidity smart contracts. Deliverables include:

- Three auditable Solidity smart contracts (World Bank Reserve, National Bank, Local Bank) implementing role-based access control, reserve management, hierarchical fund distribution, loan lifecycle management, and emergency controls.
- Integration with OpenZeppelin security libraries including `Ownable` for ownership patterns and `ReentrancyGuard` for protection against reentrancy attacks.
- Deployment and testing on EVM-compatible public test networks (Ethereum Sepolia, Polygon Mumbai).
- Gas optimization strategies and transaction economics analysis for the hierarchical lending model.
- Wallet integration via Wagmi, RainbowKit, and Viem adhering to the EIP-1193 standard.

## 12 Conclusion

The Crypto World Bank addresses a **complex, multi-party coordination and trust problem** in hierarchical development finance by leveraging the unique properties of blockchain technology: immutability, programmable enforcement, and cryptographic auditability. The solution goes beyond existing DeFi lending protocols by introducing a four-tier institutional hierarchy, AI/ML-augmented risk assessment, and a comprehensive governance framework that addresses network membership, business operations, and technology infrastructure.

The platform is positioned at the intersection of decentralized finance, institutional lending, and applied AI security—a combination that is presently unaddressed in both the academic literature and the commercial blockchain landscape. With a working prototype, a defined market and partnership strategy, transparent risk assessment, and a phased go-to-market plan, the Crypto World Bank represents a viable and scalable contribution to the emerging field of blockchain-based development finance.

## A Technology Stack

| Layer | Technology | Version / Notes |
|---|---|---|
| Smart Contract | Solidity, OpenZeppelin | 0.8.20; Ownable, ReentrancyGuard |
| Frontend | React, TypeScript, Vite, Material-UI | React 18; MUI v5; Material Design 3 theme |
| Wallet Integration | Wagmi, RainbowKit, Viem | EIP-1193 compliant |
| Build and Test | Hardhat | Automated test suite; deployment scripts |
| Target Networks | Polygon Mumbai, Ethereum Sepolia | Public testnets; zero-cost operation |
| AI/ML (planned) | Python, FastAPI, scikit-learn, SHAP | Random Forest, Isolation Forest, SHAP |

# B Smart Contract Capabilities

The platform's three smart contracts collectively support the following operations:

- **World Bank Reserve Contract:** Reserve management, deposit handling, national bank registration, fund distribution to national banks, system pause/unpause, emergency withdrawal, and global statistics.
- **National Bank Contract:** Local bank registration, borrowing from the world bank reserve, fund distribution to local banks, and network status queries.
- **Local Bank Contract:** Loan requests, loan approval and rejection, installment payments, bank user management, and approver designation.

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